

Healthcare Dataset

Exploratory Data Analysis & Insights Dashboard

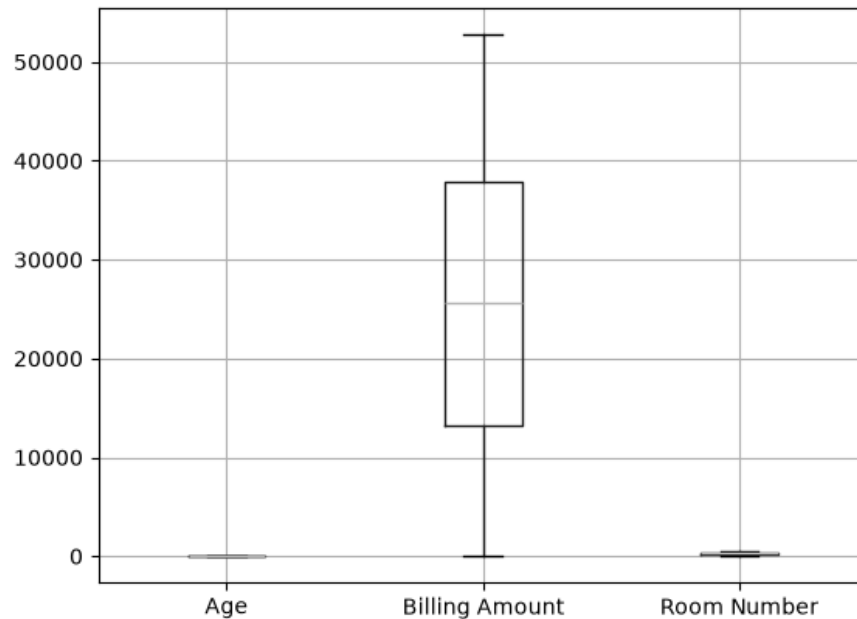
Data Cleaning • SQL Analysis • Python Visualization

Dataset	healthcare_dataset.csv
Tools	Python (Pandas, Matplotlib), SQLite (SQL)
Prepared as	Group project (midterm assessment)
Report generated	September 24, 2026

1. Data Cleaning Overview

Before analysis, the raw dataset was loaded into Pandas and SQLite. Cleaning steps included: checking for duplicate records (exact and near-duplicate patient entries), standardizing capitalization in the **Name** field, checking for stray whitespace across text columns, parsing admission and discharge dates, and validating numeric fields (Age, Billing Amount, Room Number) for out-of-range or invalid values. Records with negative Billing Amount were identified as data-entry errors and removed. The chart below summarizes the spread and outliers in the key numeric fields after cleaning.

Outlier Check — Age, Billing Amount, Room Number

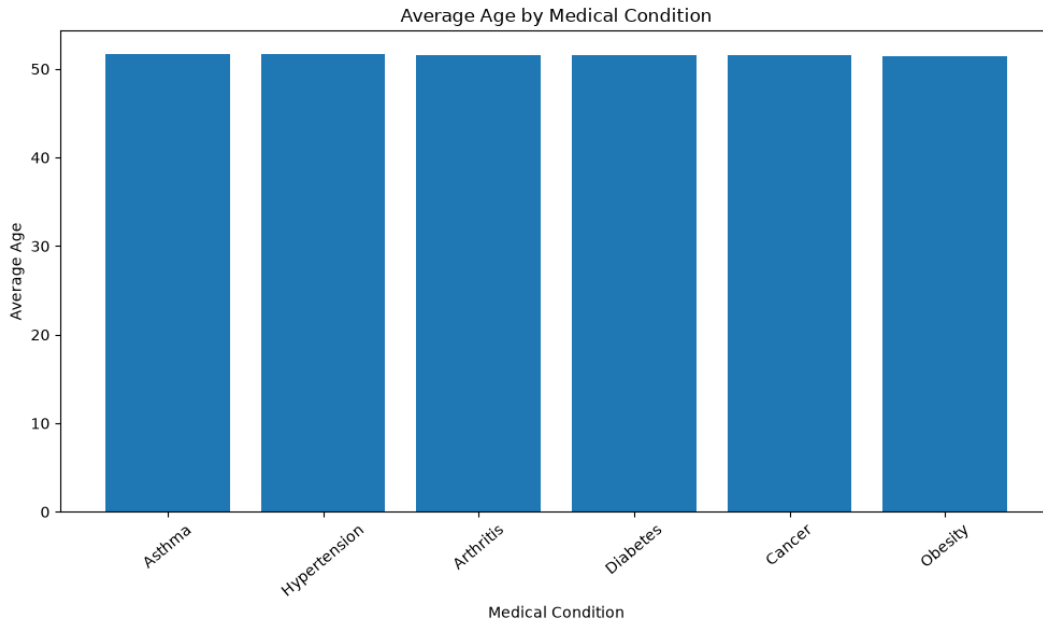


Boxplot of the three key numeric fields. Room Number and Age show a tight, expected range; Billing Amount shows a wider spread consistent with varying treatment costs across conditions.

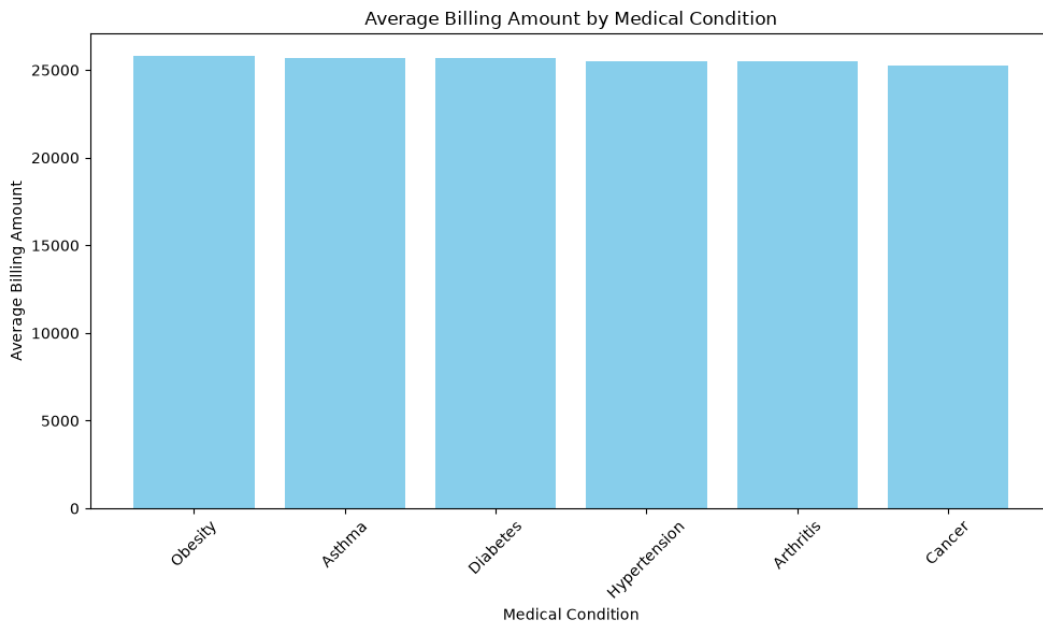
2. Age & Billing by Medical Condition

Using SQL aggregate queries against the cleaned dataset, average patient age and average billing amount were computed per medical condition to see whether certain conditions skew toward older patients or higher treatment costs.

Average Age by Medical Condition



Average Billing Amount by Medical Condition

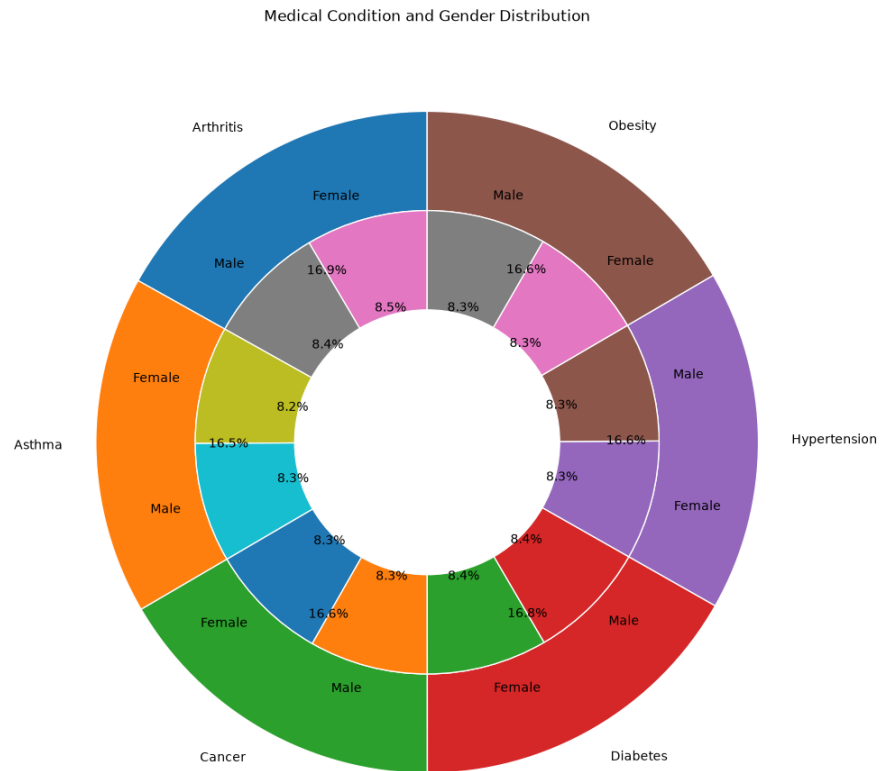


Average age is fairly close across conditions, while billing amounts vary somewhat more, reflecting differences in typical treatment intensity per condition.

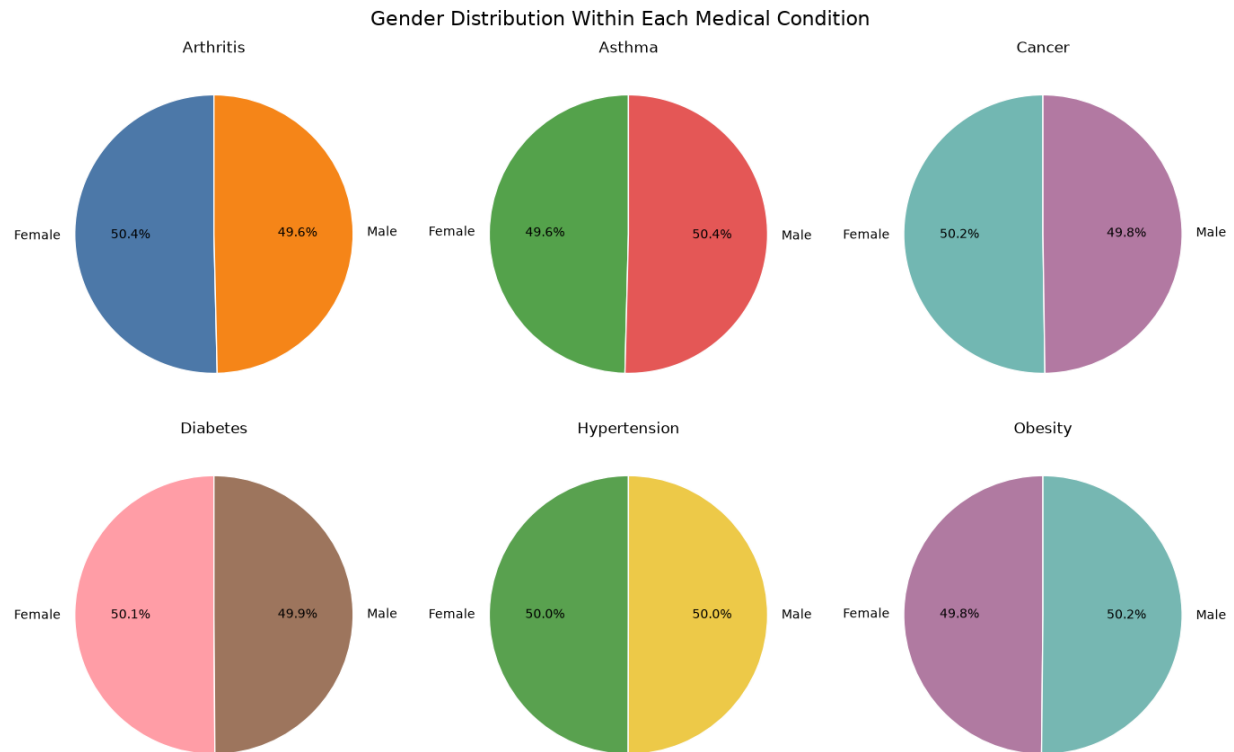
3. Gender Distribution Across Conditions

Patient counts were broken down by medical condition and gender to check whether any condition is disproportionately represented in one gender.

Medical Condition & Gender Distribution (Nested View)



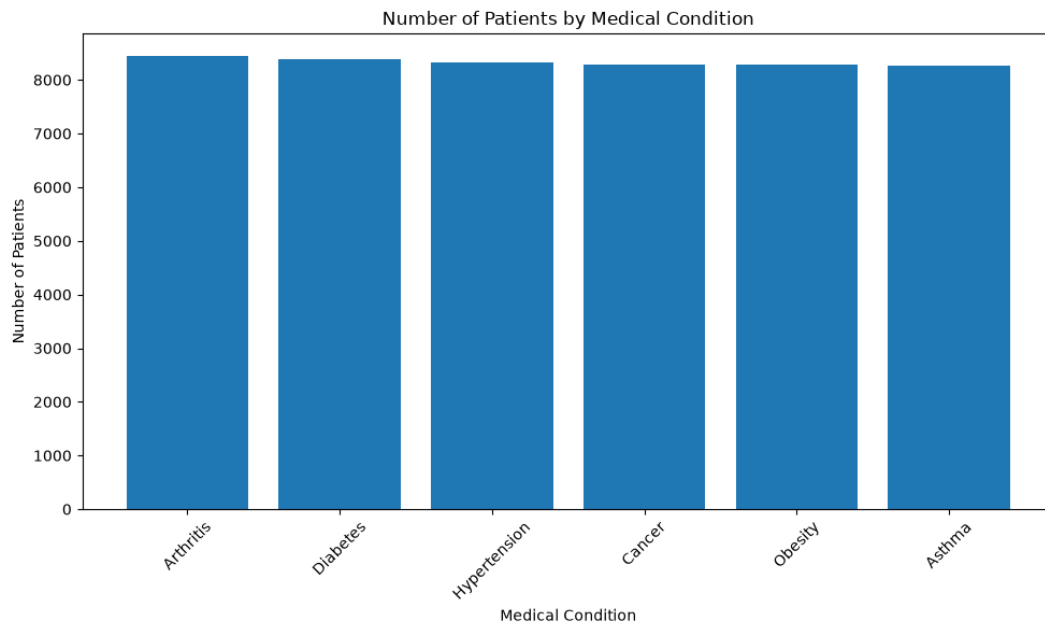
Gender Split, Per Condition



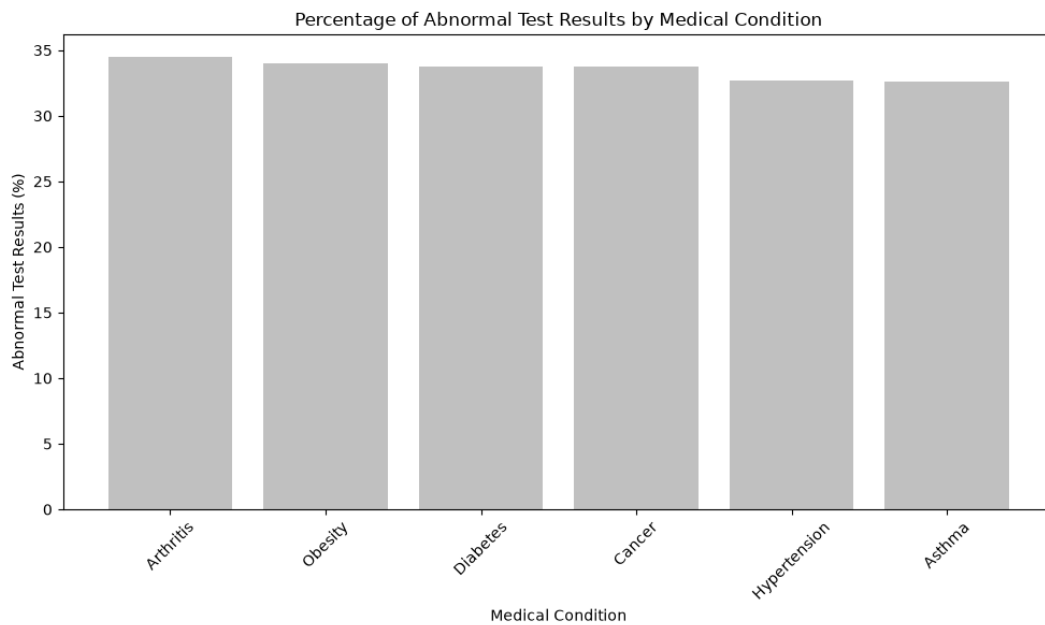
Both views show gender is close to an even 50/50 split within every medical condition in this dataset, with no condition showing a strong gender skew.

4. Patient Volume & Test Outcomes

Number of Patients by Medical Condition



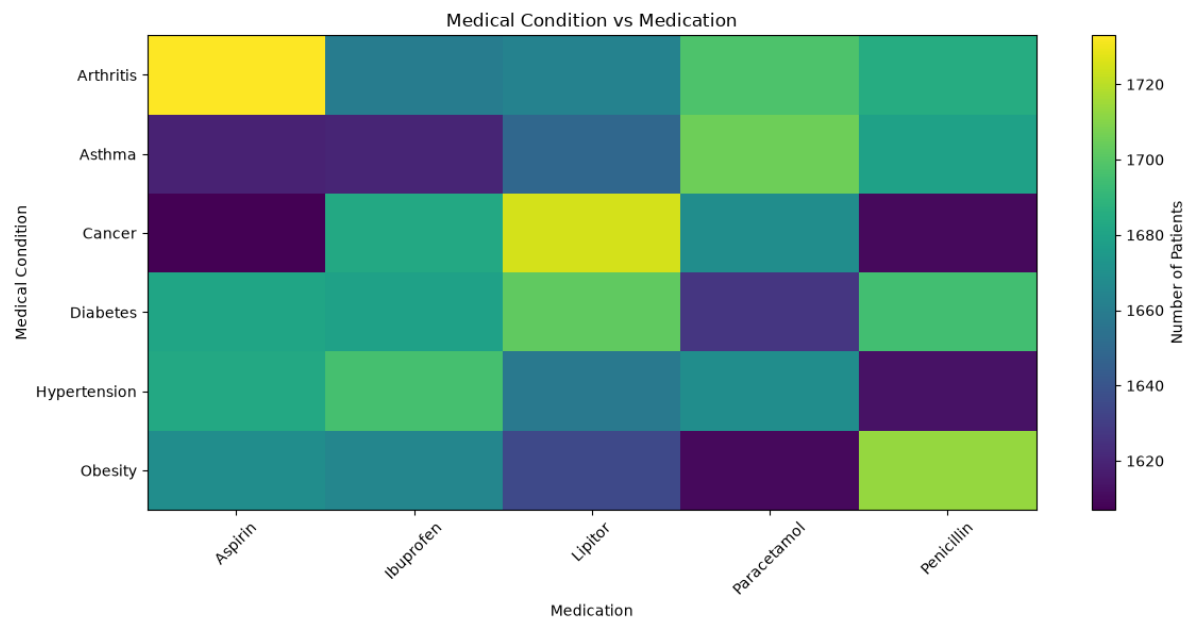
Abnormal Test Result Rate by Medical Condition



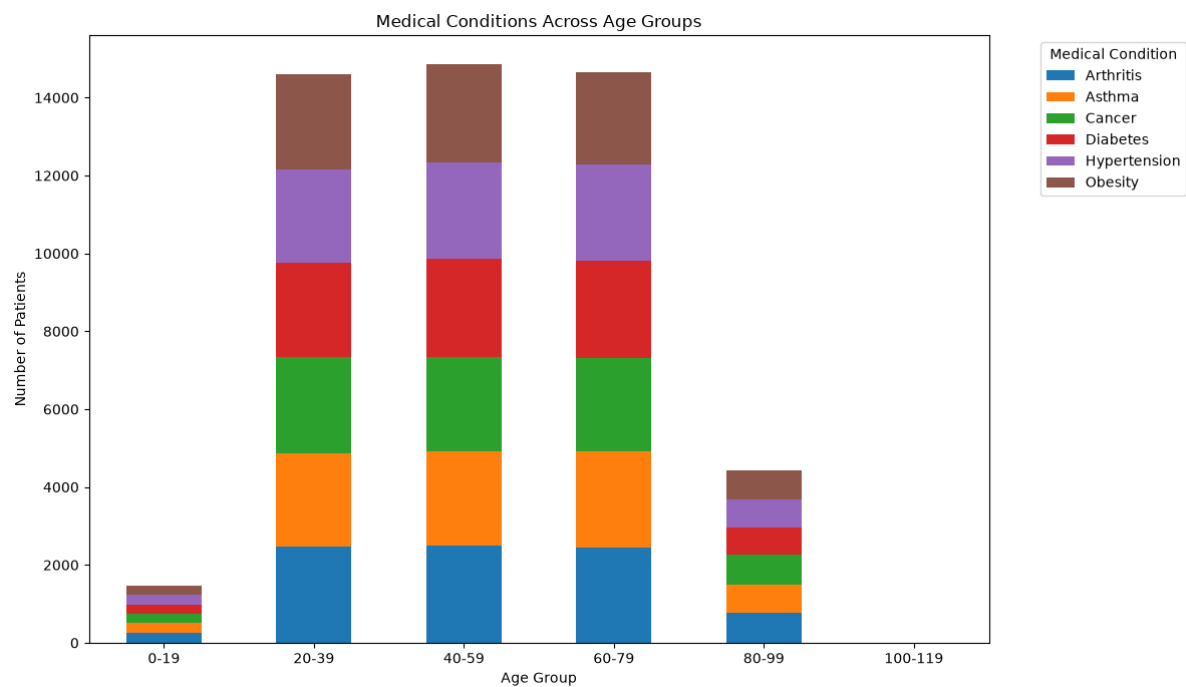
Patient counts are evenly distributed across the six conditions in the dataset. Abnormal test result rates hover close to one another as well, suggesting no single condition drives a disproportionate share of abnormal outcomes.

5. Medication Usage & Age Groups

Medication Frequency by Medical Condition



Medical Conditions Across Age Groups



The medication heatmap shows which drugs are most commonly prescribed per condition, while the stacked age-group chart shows all six conditions are represented across every adult age bracket, with the working-age and older groups carrying the largest patient volumes.